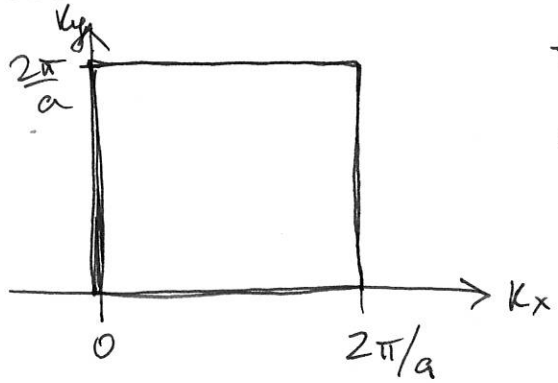
Simple 2d model of topological insulator (Atland, Fritz, U of Cologne)

Let $\hat{H} = \sum_{k \in BZ} \left(c^{+}_{k\uparrow}, c^{+}_{k\downarrow}\right) \left(\vec{\sigma} \cdot \vec{h}(k)\right) \begin{pmatrix} c_{k\uparrow} \\ c_{k\downarrow} \end{pmatrix}$

Haldane-like model

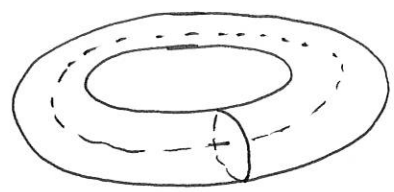
Here BZ is 2d Brillouin zone which we choose to be square $(0, \frac{2\pi}{a}) \times (0, \frac{2\pi}{a})$.



As always, opposite sides of the square are identified

$\Downarrow$

BZ is torus topologically



Let $\vec{h}(\vec{k}) = \sin(k_x a)\,\hat{x} + \sin(k_y a)\,\hat{y} + \left[r + \cos(k_x a) + \cos(k_y a)\right]\hat{z}$

Abbreviate: $k_x a = x$, $k_y a = y$, $r + \cos(x) + \cos(y) = f$.

$$\vec{h}(\vec{k}) \cdot \vec{\sigma} = \sin(x) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \sin y \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + f \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} f & \sin x - i \sin y \\ \sin x + i \sin y & -f \end{pmatrix}.$$

Energy levels: $\det\left(\vec{h} \cdot \vec{\sigma} - E\,\hat{1}\right) = 0$

$$\det \begin{pmatrix} f - E & \sin x - i \sin y \\ \sin x + i \sin y & -f - E \end{pmatrix} = 0$$

$$(E + f)(E - f) - \left(\sin^2 x + \sin^2 y\right) = 0$$

$$E^2(\vec{k}) = \sin^2 x + \sin^2 y + (r + \cos x + \cos y)^2 = |\vec{h}(\vec{k})|^2.$$

$$E_\pm(\vec{k}) = \pm\sqrt{2+r^2+2r(\cos x+\cos y)+2\cos x\cos y}$$

There are several "critical points" in this spectrum where the gap between lower (−) and upper (+) bands closes for several specific values of the parameter r:

1) at $\vec{k}=(0,0)$, $E_\pm(0) = \pm\sqrt{2+r^2+4r+2} = \pm\sqrt{(2+r)^2} = \pm|2+r|$

$\Rightarrow$ for $r=-2$ the two bands touch at $\vec{k}=\vec{0}$ point.

2) at $\vec{k}a=(\pi,\pi)$, $E_\pm(\pi,\pi) = \pm\sqrt{2+r^2-4r+2} = \pm|2-r|$

$\Rightarrow$ for $r=2$ gap closes there.

3) at $\vec{k}a=(\pi,0)$ and $(0,\pi)$:

$$E_\pm(0,\pi) = \pm\sqrt{2+r^2-2} = \pm|r|$$

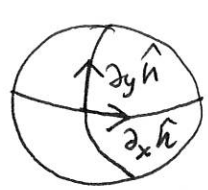
$\Rightarrow$ gap closes at $r=0$.

Next, define unit vector $\hat{n}(\vec{k}) = \dfrac{\vec{h}(\vec{k})}{|\vec{h}(\vec{k})|}$ and

ask how many times it sweeps the unit sphere as $\vec{k}\in BZ$ covers the BZ?

$$\rightarrow Q = \frac{1}{4\pi}\int_{BZ} dx\,dy\;\hat{n}(x,y)\cdot\partial_x\hat{n}(x,y)\times\partial_y\hat{n}(x,y)$$

$Q = C_1$ = Chern number



tangent vectors at point $\hat{n}(x,y)$ on the sphere

In the bulk, $E_\pm(k_x, k_y) = \pm\left(2 + r^2 + 2r(\cos k_x + \cos k_y) + 2\cos k_x \cos k_y\right)^{1/2}$

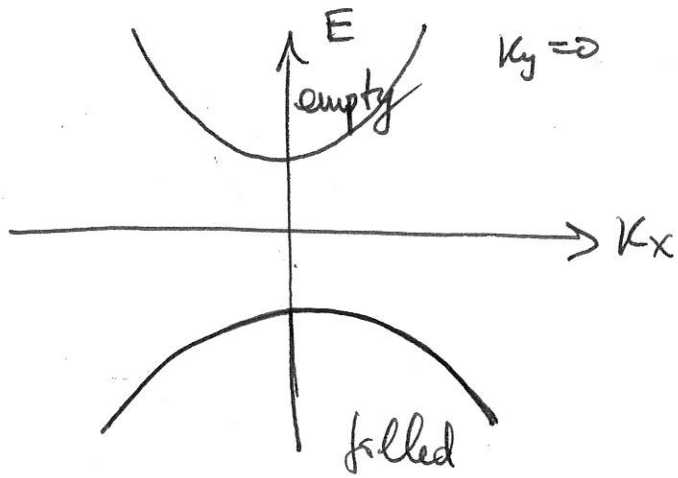
Consider $k_y = 0$ line: $E_\pm(k_x, 0) = \pm\sqrt{2 + r^2 + 2r + 2(r+1)\cos k_x}$

Near $k_x = 0$ (where gap closes at $r = -2$):

$$2 + r^2 + 2r + 2(r+1)\left(1 - \tfrac{1}{2}k_x^2\right) \cong 4 + 4r + r^2 - (r+1)k_x^2 =$$

$$= (r \cong -2) = (2+r)^2 + k_x^2$$

$$\Rightarrow E_\pm(k_x, 0) \approx \pm\sqrt{(2+r)^2 + k_x^2} = \pm\sqrt{m^2 + k_x^2}$$



Spectrum of insulator for $m \neq 0$

(the same for $m > 0$, $m < 0$.)

Example: unit vector

$$\vec{r} = (\cos\varphi \sin\theta, \sin\varphi \sin\theta, \cos\theta)$$

$$\partial_\varphi \vec{r} = (-\sin\varphi \sin\theta, \cos\varphi \sin\theta, 0)$$

$$\partial_\theta \vec{r} = (\cos\varphi \cos\theta, \sin\varphi \cos\theta, -\sin\theta)$$

Then $\vec{r} \cdot \partial_\theta \vec{r} \times \partial_\varphi \vec{r} = \sin\theta$ (ask Mathematica to do that)

and simplifies $Q = \frac{1}{4\pi} \int d\varphi\, d\theta \sin\theta$

For our case $\hat{n}(k_x, k_y) = \hat{n}(x,y) = \dfrac{\sin x\, \hat{x} + \sin y\, \hat{y} + (r + \cos x + \cos y)\hat{z}}{\sqrt{2 + r^2 + 2r(\cos x + \cos y) + 2\cos x \cos y}}$

And $\hat{n} \cdot \partial_x \hat{n} \times \partial_y \hat{n} = \dfrac{\cos x + \cos y + r \cos x \cos y}{\left(2 + r^2 + 2r(\cos x + \cos y) + 2\cos x \cos y\right)^{3/2}}$

(Mathematica file is chern-number-2d.nb)

$$\Rightarrow Q = \frac{1}{4\pi} \int_0^{2\pi} dx \int_0^{2\pi} dy \; \frac{\cos x + \cos y + r \cos x \cos y}{\left(2 + r^2 + 2r(\cos x + \cos y) + 2\cos x \cos y\right)^{3/2}}$$

depends on parameter $\underline{r}$.

Easy to integrate numerically, find $\rightarrow$

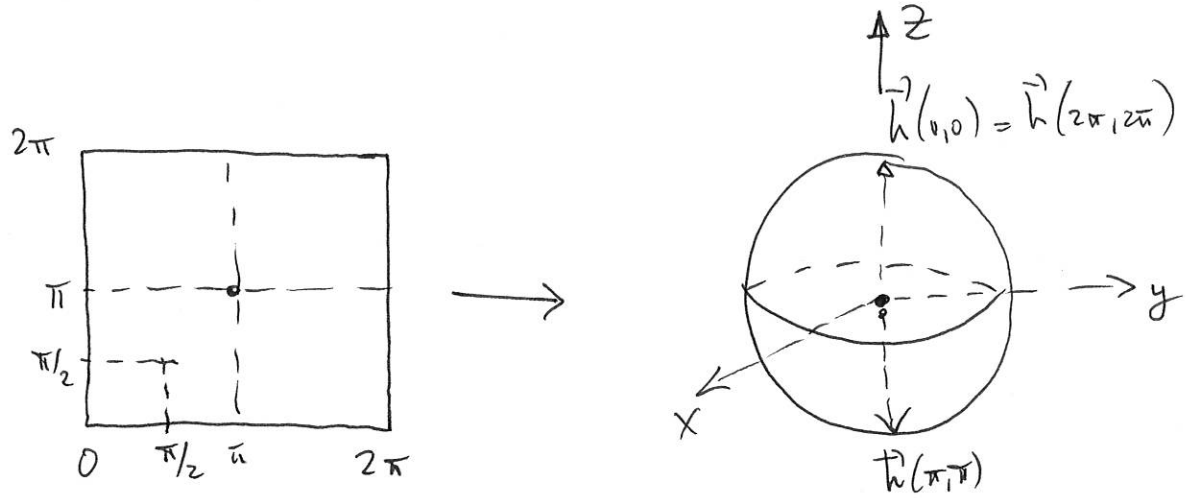
$$\begin{cases} Q = 0 \text{ for } r < -2 \\ Q = 1 \text{ for } -2 < r < 0 \\ Q = -1 \text{ for } 0 < r < 2 \\ Q = 0 \text{ for } 2 < r \end{cases}$$

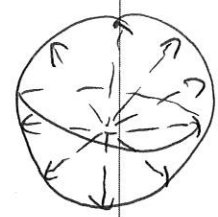
Thus, for $|r| < 2$ we have non-trivial Q!

Note: Q changes discontinuously at $r = -2$, $r = 0$, $r = 2$, which are the points where gap closes

$\Rightarrow$ topological index changes at the phase transition only! (insulator - conductor)

Consider $\underline{-2 < r < 0}$





$$\begin{cases} \vec{h}(\pi,\pi) = (r-2)\hat{z} & \text{points at south pole.} \\ \vec{h}(0,0) = (r+2)\hat{z} & \text{points at North pole. (!)} \\ \vec{h}\left(\frac{\pi}{2},\frac{\pi}{2}\right) = \hat{x} + \hat{y} + r\,\hat{z} & \end{cases}$$

As $\vec{k}$ increases from (π,π), $\hat{n} = \frac{\vec{h}}{|\vec{h}|}$ covers the sphere from the South to the North pole.

$\Rightarrow$ $Q = 1$ indeed

Now, let $r > 2$: easy to see that z-component of $\vec{h}$ $(= r + \cos(x) + \cos(y))$ ~~[illegible]~~ is always positive! $\Rightarrow$ vector $\hat{n}$ does not cover the full sphere

$\Rightarrow$ $Q = 0$.

Analytical progress: focus on the neighborhood of $r=-2$ point

$$\left.\begin{array}{ll} Q=0 & \text{for } r<-2 \\ Q=1 & \text{for } r>-2 \end{array}\right\} \Rightarrow$$ we also know that the gap closes ~~at~~ at $\vec{k}=(0,0)$

$\Rightarrow$ consider small-$\vec{k}$ approximation to $\vec{h}$:

$$\vec{h} \approx x\hat{x} + y\hat{y} + \underbrace{(r+2)}_{m}\hat{z} \qquad (\text{denote } m \equiv r+2)$$

$$\hat{n} = \frac{x\hat{x} + y\hat{y} + m\hat{z}}{\sqrt{x^2+y^2+m^2}}$$

Then we find $\hat{n}\cdot\partial_x\hat{n}\times\partial_y\hat{n} = \dfrac{m}{(x^2+y^2+m^2)^{3/2}}$

Let us now split Q integral into $Q_<$ and $Q_>$:

$$Q = Q_< + Q_>$$

$$Q_< = \frac{1}{4\pi}\int_{\Omega'} dx\,dy\;\hat{n}\cdot\partial_x\hat{n} * \partial_y\hat{n}$$

(Ω' small neighborhood of $\vec{k}=\vec{0}$ where $|\vec{k}|\ll 1$ approximation works)

$$Q_> = \frac{1}{4\pi}\int_{\text{Sphere}-\Omega'} dx\,dy\;\hat{n}\cdot\partial_x\hat{n}\times\partial_y\hat{n}$$

$\leftarrow$ rest of the BZ where $\vec{h}$ can not be approximated by ~~the~~ such expansion.

$$Q_< = \frac{1}{4\pi} \int_{\Omega'} dx\,dy \, \frac{m}{(x^2+y^2+m^2)^{3/2}} =$$

$$= \frac{1}{4\pi} \int_0^{\sqrt{\Lambda'}} dr \cdot r \cdot 2\pi \, \frac{m}{(r^2+m^2)^{3/2}} = (r^2 = u)$$

$$= \frac{\pi m}{4\pi} \int_0^{\Lambda'} \frac{du}{(u+m^2)^{3/2}} = \frac{\pi m}{4\pi} \left(- \frac{2}{(u+m^2)^{1/2}} \right) \Bigg|_0^{\Lambda'}$$

Λ' is some upper-cutoff of order 1, but since the integral is convergent, we can push it to ∞: $\Lambda' = \infty$

$$\Rightarrow \quad Q_< = \frac{-2\pi m}{4\pi} \frac{1}{(u+m^2)^{1/2}} \Bigg|_0^{\infty} = \frac{1}{2} \frac{m}{\sqrt{m^2}}$$

$$Q_< = \frac{1}{2} \operatorname{sgn}(m) = \begin{cases} -\frac{1}{2} & \text{for } r < -2 \\ +\frac{1}{2} & \text{for } r > -2 \end{cases}$$ discontinuous!

In calculating $Q_>$ we need to use full function h. But it is far from the "critical" region $\vec{k} \approx (0,0)$

$\Rightarrow$ can assume $Q_>$ does not change across $\underline{r=-2}$ transition.

$\Rightarrow$ fix $Q_>$ by requiring that full Q at $r<-2$ coincides with numerically calculated $\Rightarrow$

$Q = 0$ for $r < -2 \Rightarrow Q_> = +\frac{1}{2}$ for $r < -2$.

$\Rightarrow$ for $r > -2$ we find

$$Q = \underset{\left(+\frac{1}{2}\right)}{Q_<} + \underset{\left(+\frac{1}{2}\right)}{Q_>} = +1 \; (!)$$

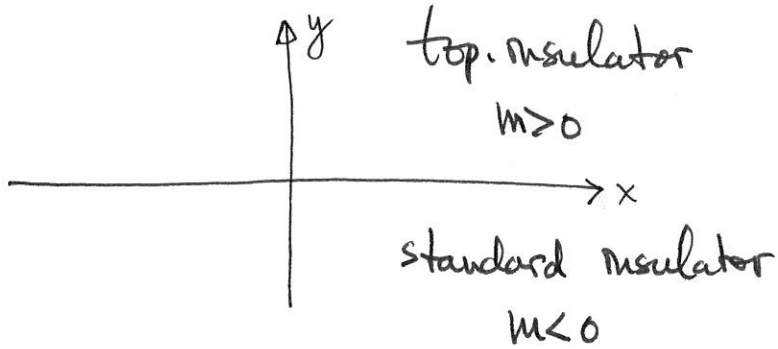
$\Rightarrow$ in this way we see that discontinuity in Q comes from the gap-closing region of the band structure!

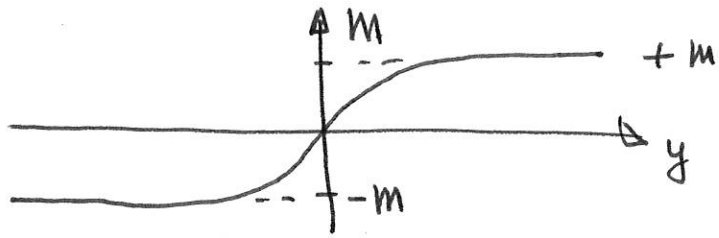
<u>Boundary mode</u>: consider again vicinity of $r=-2$.

let $m = r+2$

so that for $m<0$ $Q=0$ (trivial insulator)

for $m>0$ $Q=+1$ (topological insul.)





Assume gap $m(y)$ has some profile.

What is important is that $m(y) \to \begin{cases} +m, & y \to +\infty \\ -m, & y \to -\infty \end{cases}$

Is there solution confined to the boundary?

Note that in the bulk (i.e. for $|y| \to \infty$) all excitations are gapped because we have band insulators on both sides of the divide.

$$\hat{H} \cong \hat{p}_x \sigma^x + \hat{p}_y \sigma^y + m\sigma^z$$

$\hat{H}\psi = E\psi \Rightarrow$ let's try $\psi(x,y) = e^{ip_x x}\varphi(y)$

$$p_x \underbrace{\sigma^x}_{e^{ip_x x}\varphi(y)} - i\frac{d}{dy}\left(\sigma^y e^{ip_x x}\varphi(y)\right) + m\overset{\sigma^z}{\checkmark} e^{ip_x x}\varphi(y) = E\, e^{ip_x x}\varphi(y)$$

Start with particular case of $p_x = 0$ and look for a zero-energy ($E=0$) solution:

$$-i\sigma^y \frac{d\varphi}{dy} + m(y)\sigma^z\varphi = 0; \quad \varphi(y) = \begin{pmatrix} u(y) \\ v(y) \end{pmatrix}$$

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} u' \\ v' \end{pmatrix} + m\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = 0$$

$$\begin{cases} -v' + m(y)u = 0 \\ u' - m(y)v = 0 \end{cases}$$

let $u = e^{\lambda(y)}u_0$, $v = e^{\lambda(y)}v_0$

$$\frac{de^{\lambda}}{dy} = e^{\lambda}\frac{d\lambda}{dy} = e^{\lambda}\lambda'$$

$$\Rightarrow \begin{cases} -\lambda' v_0 + m(y)u_0 = 0 \\ \lambda' u_0 - m(y)v_0 = 0 \end{cases}$$

$$\begin{aligned} \lambda' u_0 &= m(y)v_0 \\ \lambda' v_0 &= m(y)u_0 \end{aligned} \Rightarrow \frac{u_0}{v_0} = \frac{v_0}{u_0} \Rightarrow u_0^2 = v_0^2$$

$$u_0 = \pm v_0$$

a) Suppose $u_0 = v_0 \Rightarrow \lambda' = m(y)$

$$\frac{d\lambda}{dy} = m(y)$$

$$\int_0^y \frac{d\lambda}{dy'} dy' = \int_0^y m(y') dy'$$

$$\lambda(y) - \lambda(0) = \int_0^y m(y') dy'$$

Since $m(y') \xrightarrow[y' \to \infty]{} +m$ (so that we indeed have top. insulator for $y > 0$)

$$\Rightarrow \int_0^y m(y') dy' \approx my > 0 \qquad \text{(diverges for } y \to +\infty)$$

$$\Rightarrow \varphi(y) \sim e^{my} \begin{pmatrix} u_0 \\ v_0 \end{pmatrix} \text{ is NOT normalizable!}$$

b) Suppose $u_0 = -v_0 \Rightarrow -\lambda'(-u_0) + m(y) u_0 = 0$

$$\frac{d\lambda}{dy} = -m(y)$$

$$\Rightarrow \lambda(y) - \lambda(0) = -\int_0^y m(y') dy' \to -my$$

Then $\varphi(y) \sim e^{-my} u_0 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ <u>is normalizable.</u>

Thus, search for $\psi(x,y) = e^{ip_x x} e^{\lambda(y)} u_0 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$$\left[p_x \sigma^x - i \sigma^y \frac{d\lambda}{dy} + m(y) \sigma^z \right] e^{ip_x x} e^{\lambda(y)} \begin{pmatrix} 1 \\ -1 \end{pmatrix} u_0 =$$

$$= E\, e^{ip_x x} e^{\lambda(y)} \begin{pmatrix} 1 \\ -1 \end{pmatrix} u_0$$

$$\left[p_x \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \lambda' + m \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right] \begin{pmatrix} 1 \\ -1 \end{pmatrix} = E \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$p_x \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \lambda' \begin{pmatrix} 1 \\ 1 \end{pmatrix} + m \begin{pmatrix} 1 \\ 1 \end{pmatrix} = E \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{cases} -p_x + \lambda' + m = E \\ p_x + \lambda' + m = -E \end{cases} \Rightarrow \begin{cases} +\lambda' + m = E + p_x \\ \lambda' + m = -E - p_x \end{cases}$$

$$\Rightarrow E + p_x = -E - p_x \Rightarrow 2E = -2p_x$$

$$\Rightarrow E = -p_x$$

$\Rightarrow$ we get back $\dfrac{d\lambda}{dy} = -m(y)$

$$\lambda(y) = -\int_0^y m(y')\,dy'$$

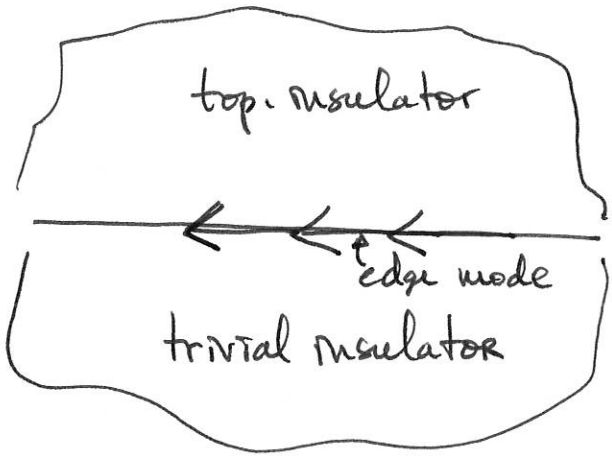
So, $\psi(x,y) = e^{ip_x x} e^{-\int_0^y m(y')dy'} \begin{pmatrix} 1 \\ -1 \end{pmatrix} u_0$ is the solution: it is localized near the boundary.

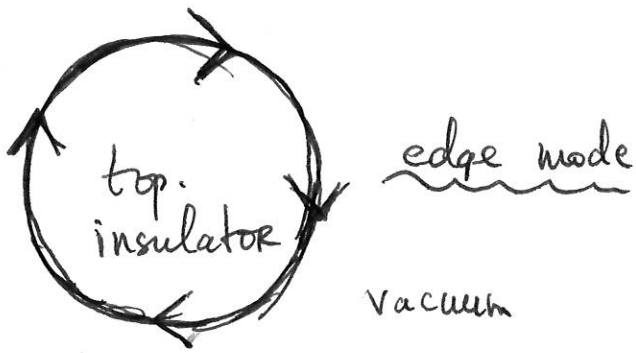
$E = -v_F p_x$

inserted by dimensional argument (was 1 before)

Group velocity of the boundary mode

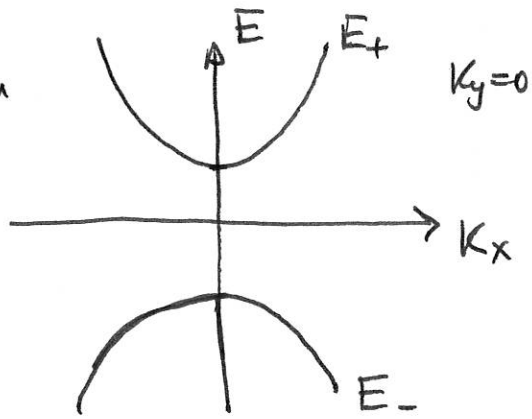
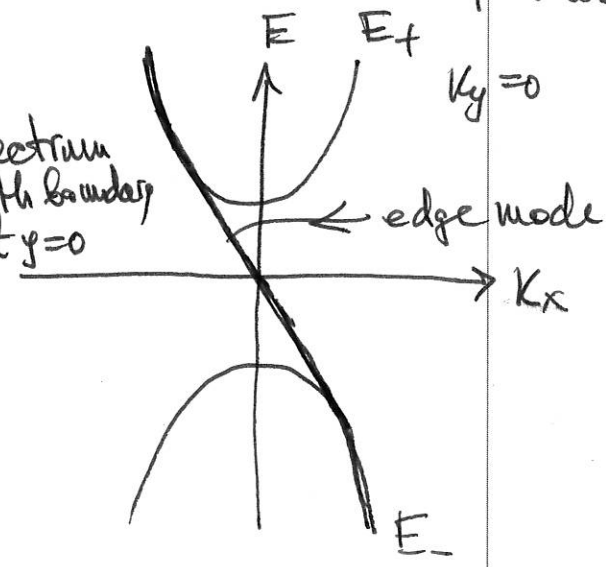
$$\frac{dE}{dp_x} = -v_F \Rightarrow \text{this is } \underline{\text{left-moving}} \text{ (chiral) } \underline{\text{edge mode}}$$





Top. insulator (with $Q \neq 0$) cannot be smoothly connected with trivial insulator, ($Q=0$), or vacuum: gapless edge mode must be present!

Spectrum of top. insulator without boundary



Spectrum with boundary at $y=0$